

Physics of amorphous silicon based alloy field-effect transistors

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In this paper we develop a new theory to describe the characteristics of amorphous silicon based alloy field-effect transistors. We show that the transition from below to above threshold operation occurs when the Fermi level in the accumulation region moves from the deep to tail localized states in the energy gap. The current-voltage and capacitance-voltage characteristics are related to the basic material parameters such as the distribution of localized states in the energy gap, band mobility, device geometry, channel doping, and series resistances. Our analysis shows that an on current in excess of 2×10^{-7} A/ μ m gate width can be obtained with a 10- μ m gate length. We also demonstrate that even in the above threshold regime the field-effect mobility is dependent on the gate voltage. Our theory can be used to optimize the design of amorphous silicon based alloy field-effect transistors.

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INTRODUCTION

Amorphous silicon based alloy (*a*-Si) field-effect transistors have emerged as a viable and important technology for large area, low cost integrated circuits. These circuits are currently being used to drive large area liquid crystal displays and have good potential for basic integrated circuits (IC's) and addressable image sensing arrays.

Possible improvements in *a*-Si technology depend on an understanding of the device physics. In this paper we propose a new physical mechanism describing the *a*-Si field-effect transistor (FET) operation and develop it into a theory describing their operation. We also check the validity of our analytical approach using a computer simulation based on the solution of the complete set of transport equations for an *a*-Si FET and demonstrate that our theory is in good agreement with experimental results. An extended summary of this work has been published in the Proceedings of the 10th International Conference on Amorphous and Liquid Semiconductors.¹

In this paper we introduce two basic operational modes of an *a*-Si thin film transistor (TFT). Close to the flat band condition nearly all the charge induced in the *a*-Si channel goes into deep localized states in the energy gap. With an increase in gate voltage more states are filled by a larger induced charge, so that the Fermi level at the insulator-amorphous silicon interface moves closer to the conduction band. This leads to an increase in the concentration of mobile carriers in the conduction band which rises superlinearly with the gate voltage. We will refer to this mode of operation as the subthreshold or below threshold regime.

In crystalline field-effect transistors above threshold nearly all the charge induced in the channel appears in the conduction band. This will happen when virtually all localized states are filled and the Fermi level at the surface of the conducting channel is within $\sim kT/q$ of the conduction band edge. As our calculation shows this condition would require a very high electric field in the insulator (in excess of 2×10^6 V/cm), i.e., high gate voltages. However, a dramatic change

in the device *I-V* characteristics does indeed occur at substantially lower field strengths and gate voltages. As discussed below, this may be explained by considering in detail the energy distribution of the localized states in amorphous silicon.

Photoconductivity² and field-effect data³⁻⁵ show that the localized acceptor-like states (which are important for the *n*-channel device operation) may be crudely divided into two groups: deep localized acceptor states and "tail" states (see Fig. 1) with the transition between them occurring 150–200 mV below the conduction band edge. The important role played by the tail states in determining the above threshold conductance in the channel of an amorphous silicon TFT was first discussed by Powell.⁶⁻⁸ In his computer simulation he assumed a constant density of deep states and a linear distribution of tail states.

Based on the above experimental results²⁻⁶ we assume that the density of acceptor-like states $g_A(E)$ variation for both types of states is approximately exponential with energy:

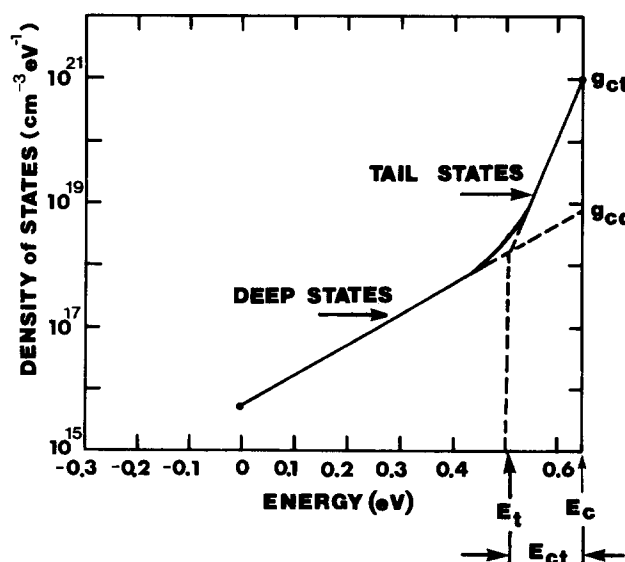


FIG. 1. Distribution of localized acceptor states in the energy gap of amorphous silicon.

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$$g_A(E) = g_1 \exp\left(\frac{E}{E_1}\right) + g_2 \exp\left(\frac{E}{E_2}\right), \quad (1)$$

where $E_{1,2} = kT_{1,2}$. For the tail states the characteristic temperature T_1 of this variation is about 300 K, whereas for the deep states the characteristic temperature T_2 is around 1000 K.

When the Fermi level is in the energy range corresponding to the deep localized states, most of the charge is located close to the Fermi level. However, if T_1 is less than or equal to T (the lattice temperature) and the Fermi level is in the tail states, the product of the density of states and the Fermi-Dirac function increases with increasing energy, i.e., as the Fermi level moves closer to the conduction band edge.

Consequently, when the Fermi level is in the tail states, most of the charge is actually located above the Fermi level because the characteristic energy kT_1 of the exponential variation of the density of tail states is close to the thermal energy at room temperature ($T_1 \sim 300$ K). This is illustrated by Fig. 2, which shows the normalized product of the density of states and the Fermi-Dirac distribution function for the exponential density of states distributions for varying ratios T_1/T .

As a consequence of such charge distribution the shift of the Fermi level with gate voltage in this regime is much smaller than in the subthreshold regime. Also, the ratio of the mobile to localized charge is less sensitive to the shift of the Fermi level as the charge distribution in the localized tail states is less dependent on the Fermi level position (see Sec. II). This is equivalent to saying that when the Fermi level is in the tail states the effective field-effect mobility becomes much less sensitive to the gate voltage and temperature than in the subthreshold regime. We refer to this regime as an "above threshold" regime. Moreover, as will be shown in Sec. VII, the current-voltage characteristics in this regime look similar to the above threshold voltage characteristics of a conventional crystalline MOSFET, although the field-effect mobility is several orders of magnitude smaller than in crystalline devices and is also dependent on the gate voltage

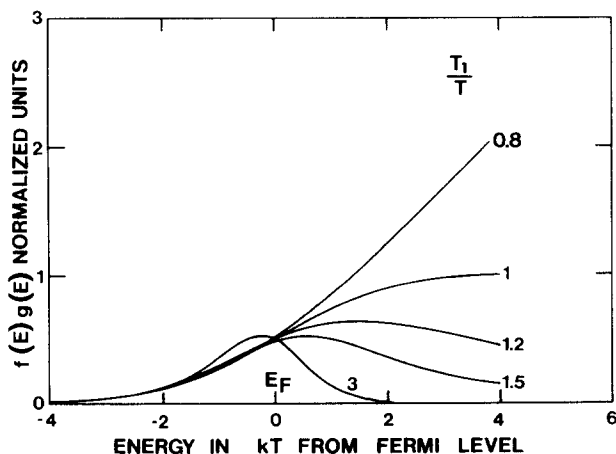


FIG. 2. Plot of the product of the Fermi-Dirac function and an exponential density of states (normalized to a half at the Fermi level) as a function of energy for varying characteristic temperatures of the density of states distribution.

and temperature. Depending on the induced charge and the density of the interface states the field-effect mobility may vary from around one percent to a significant fraction of the band mobility. In a way, the above threshold regime in *a*-Si TFT's is intermediate between the below threshold regime and the regime similar to the above threshold regime in conventional crystalline MOSFETs. However, as mentioned above and as demonstrated in this paper, the latter regime may only occur if the electric field strength across the gate insulator is in excess of 2×10^6 V/cm, whereas the above threshold regime we discuss only requires a field across the gate $\sim 2 \times 10^5$ V/cm.

Our analysis of the *a*-Si FET starts from a calculation of the shift of the Fermi level in the channel as a function of the induced charge which is directly related to the gate voltage. The induced charge is divided between both the deep and tail localized states as well as mobile carriers. We calculate the localized charges as functions of Fermi level position analytically by integrating the density of states spectrum with the Fermi-Dirac statistics. Our more accurate computer simulation is based on a numerical integration which takes into account the occupancy of the localized states predicted by the Shockley-Read-Hall statistics. It is in good agreement with our analytical treatment.

Based on the calculated band bending we then derive equations describing the current voltage (*I-V*) and capacitance voltage (*C-V*) characteristics of an *a*-Si TFT in the below and above threshold regimes using the gradual channel approximation.⁹ We also analyze the temperature and doping dependences of these characteristics and show that the device characteristics are significantly altered by the source and drain series resistances. The importance of the source and drain series resistances in limiting device performance was first discussed by Matsumura *et al.*¹⁰ We demonstrate that the characteristic energy slope E_2 of the density of deep localized acceptor states may be deduced from the experimental *I-V* characteristics in the subthreshold regime. In good agreement with other techniques we deduce $E_2 \sim 80$ – 100 mV using the experimental data reported by different researchers.

Our computer calculation of the drain-to-source conductance as a function of the induced charge in the channel is in good agreement with our analytical theory for both the below and above threshold regimes. We also discuss how additional interface localized states and band bending at the *a*-Si interface away from the insulator⁸ may affect our results.

Our theory of *a*-Si thin film transistors predicts how device characteristics scale with gate length, insulator thickness, channel doping, temperature, and material properties. This theory of the steady-state operation may be used to characterize devices and to optimize the design of *a*-Si FETs.

II. DENSITY OF LOCALIZED STATES IN THE ENERGY GAP OF *a*-Si

Current-voltage and capacitance-voltage characteristics of *a*-Si TFT's strongly depend on the density and distribution of localized states in the energy gap. The position of the carrier Fermi level in an *a*-Si channel may vary from

energies close to the bottom of the conduction band to energies near to the top of the valence band depending on the sign and magnitude of the gate voltage. The shift of Fermi level with gate bias in a TFT is larger than the shift of the quasi-Fermi levels in *a*-Si solar cells under AM1.5 illumination at 300 K.

In an amorphous silicon *p-i-n* solar cell operating under global AM1.5 illumination at 300 K, the quasi-Fermi levels do not enter the tail states^{11,12} and, hence, a simple model which describes the density of states spectrum as the sum of two exponentially varying donor and acceptor localized states is adequate to describe their operation. However, as we show below in *a*-Si TFT's the Fermi level does indeed enter the tail states and the above simple model of the density of states spectrum is no longer adequate.

The acceptor-like localized states will determine the characteristics of *n*-channel devices, whereas donor-like states will dominate the characteristics of *p*-channel devices. The acceptor-like localized states are mostly located in the upper part of the energy gap and the donor-like states are mostly located in the lower part of the energy gap. A realistic distribution of localized acceptor states for *a*-Si is shown in Fig. 1. It consists of exponentially varying distributions of both deep and tail states. The slopes of the distribution may be found by different techniques such as transient and steady-state photoconductivity,² etc. For the donor-like states the information about the slopes may be obtained from the intensity dependence of the minority carrier diffusion length.¹³ The characteristic slope of the conduction band tail states is approximately the thermal energy at room temperature, and this has very important consequences which will be discussed in more detail in Sec. VI.

III. CHARGE IN LOCALIZED STATES

As will be shown in Sec. IV, most of the charge induced into the channel of an *a*-Si TFT goes into the localized or tail states depending on the applied gate voltage. The density of this charge is determined by the density of the localized tail states and by the position of the carrier Fermi level.

The density qN of charge induced into both the deep localized and tail states is given by

$$qN = \int_{E_{F0}}^{E_c} \frac{g_A(E) dE}{1 + \exp\left(\frac{E - E_F}{kT}\right)}, \quad (2)$$

where q is the electronic charge, g_A the density of localized acceptor states, E the energy, E_{F0} the bulk Fermi level, E_F the electron Fermi level, and E_c the bottom of the conduction band.

Strictly speaking, the occupation of the localized states is determined by the trap quasi-Fermi levels. However, a detailed analysis¹⁴ shows that the carrier Fermi level is very close to the trap quasi-Fermi level for the acceptor states. Also, for simplicity we have assumed that in Eq. (1) the g factor of the acceptor states is equal to unity.

In the case when the shift of the electron Fermi level under the gate bias is larger than both the thermal energy kT and the characteristic energy of the exponential variation of the acceptor states. Eq. (2) may be rewritten as

$$N_{loc} = g_c \frac{kT_F}{q} \exp\left(\frac{-E_{cF}}{kT_F}\right) \int_0^{\exp\left(\frac{E_{cF}}{kT_F}\right)} \frac{dz}{1 + z^\alpha}, \quad (3)$$

where we have substituted

$$z = \exp\left(\frac{E - E_F}{kT_F}\right),$$

or

$$N_{loc} = \frac{g_c kT_F}{q} \exp\left(\frac{-E_{cF}}{kT_F}\right) J(\alpha), \quad (4)$$

where

$$E_{cF} = E_c - E_F, \quad (5a)$$

and

$$J(\alpha) = \int_0^{\exp\left(\frac{E_{cF}}{kT_F}\right)} \frac{dz}{1 + z^\alpha} \quad (5b)$$

Here, kT_F is the characteristic energy of the exponential variation of the density of states in the vicinity of the electron Fermi level,

$$\alpha = T_F/T, \quad (6)$$

and g_c is the density of the localized states extrapolated to the bottom of the conduction band assuming that T_F is independent of energy. We have assumed that the acceptor states consist of two distinct exponential regions and hence both g_c and T_F are different in each range. In other words, when the Fermi level is in the energy range corresponding to the tail states $g_c = g_{ct}$ and when the electron Fermi level is in the energy range corresponding to the deep localized states $g_c = g_{cd}$ (see Fig. 1).

The integral J in Eq. (4) is determined by the parameter α . For the electron Fermi level located in the deep localized states $T_F = 1000$ K and $\alpha \simeq 3.3$ is greater than unity at room temperature or below. For the tail states we assume $T_F \simeq 260$ K because $T_1 < T = 300$ K better illustrates the above threshold regime. The exact value of T_F for the tail states is not known very accurately and, as we show below, a variation in T_1 of 40 K from 260 K only changes the field-effect mobility by about 50%. Hence, for the tail states α may be taken to be less than or equal to unity.

For the case $\alpha > 1$, $T_F = T_2$, and Eq. (3) may be rewritten as

$$qN_{deep} = g_c kT_2 \exp\left(\frac{-E_{cF}}{kT_2}\right) \int_0^\infty \frac{dz}{1 + z^\alpha}, \quad (7)$$

to yield

$$qN_{deep} = \frac{g_F \pi kT}{\sin(\pi T/T_2)}, \quad (8)$$

where

$$g_F = g_{cd} \exp\left(\frac{-E_{cF}}{kT_2}\right), \quad (9)$$

is the density of the deep localized states at the electron Fermi level. Equation (8) is in agreement with the calculation of the localized charge density given in Ref. 15.

For the tail states ($\alpha < 1$) the integral J in Eq. (4) has to be evaluated numerically. However, it may be still evaluated analytically for $\alpha = 0$ and for $\alpha = 1$:

$$J(\alpha = 0) = \frac{1}{2} \exp(E_{cF}/kT_1), \quad (10)$$

$$J(\alpha = 1) = \ln \left[\exp\left(\frac{E_{cF}}{kT_1}\right) + 1 \right]. \quad (11)$$

We have found that a suitable analytic interpolation covering the range $0 < \alpha < 1$ is given by

$$J(0 < \alpha < 1) = \frac{1}{2} \exp(a) \left\{ \frac{2 \ln[\exp(a) + 1]}{\exp(a)} \right\}^\alpha, \quad (12)$$

where $a = E_{cF}/kT_F$.

Figure 3 gives a comparison between the numerical integration for J and the interpolation as given by Eq. (12) as a function of α for three different values of E_{cF}/kT_F . In most practical cases, E_{cF} will be greater than $2kT_1$ and hence Eq. (12) can be simplified to

$$J(0 < \alpha < 1) = \frac{1}{2} \exp(a) \left[\frac{2a}{\exp(a)} \right]^\alpha, \quad (13)$$

leading to the following expression for the localized charge in the tail states:

$$qN_{\text{tail}} = \frac{1}{2} g_{ct} kT_1 (2a)^\alpha \exp\left(\frac{-E_{cF}}{kT}\right). \quad (14)$$

As can be shown based on the comparative analysis of Eqs. (8) and (14), there is an important qualitative difference between the case when the electron Fermi level is in the deep localized state and the case when the electron Fermi level is pushed into the tail state. In the former case which corresponds to relatively small band bending, i.e., to relatively small gate voltages, the localized charge is more or less located within an energy $kT_F = kT_2$ below the electron Fermi level. In the latter case which is achieved when the gate voltage (and as a result the band bending) is relatively large, most of the localized charge is distributed above the electron Fermi level. In the limiting case $T_1/T \ll 1$ most of the localized charge is actually located within kT_1 of the bottom of the conduction band.

As shown in Sec. VII, the ratio of the mobile charge to localized charge in this above threshold regime is much less dependent on the gate voltage than in the subthreshold re-

gime, and the surface density of free carriers induced by the gate voltage is very crudely proportional to the voltage difference between the gate and the channel to the power of 1.5.

The important difference, however, is that most of the induced charge is still in the localized states. Nevertheless, the device characteristics look like MOSFET characteristics with a low effective electron mobility. We refer to this regime as the "above threshold" regime and distinguish it from the "below threshold" regime when the electron Fermi level is in the deep localized states and, as shown in Sec. IV, the induced charge of the free electrons is small and increases sharply with an increase of gate voltage.

The total density N_{loc} of the localized charge may be approximated by the following expression:

$$N_{\text{loc}} = N_{\text{deep}} + N_{\text{tail}}. \quad (15)$$

In Fig. 4 we show N_{deep} , N_{tail} , N_{loc} , and n_{free} where

$$n_{\text{free}} = N_c \exp\left(\frac{E_F - E_c}{kT}\right), \quad (16)$$

as functions of the Fermi level position. This figure clearly illustrates the distribution of the induced charge amongst both the deep and tail localized states as well as the conduction band.

The results shown in Fig. 4 may be directly compared with the dependence of the induced localized and mobile charges on the position of the Fermi level calculated in Ref. 15 (see Fig. 5) where both the role played by the tail states has been neglected and T_2 was assumed to be ~ 1800 K. As can be seen from Fig. 5, the theory developed in Ref. 15 predicts that the mobile charge actually becomes larger than the localized charge for $E_c - E_F \simeq 0.33$ V. When the Fermi level is moved even closer to the conduction band the field-effect mobility according to the results of Ref. 15 would be very close to the band mobility. In contrast, however, our model predicts that even in the above threshold regime described above the field-effect mobility is only a fraction of the band mobility as the mobile induced charge always remains less than the localized induced charge.

Another important difference between our model and

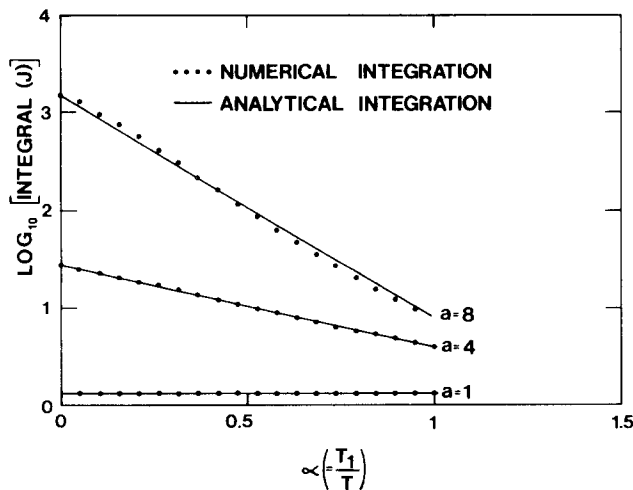


FIG. 3. Comparison between the numerical calculation of the integral J (crosses) and the interpolation formula (8) (solid line).

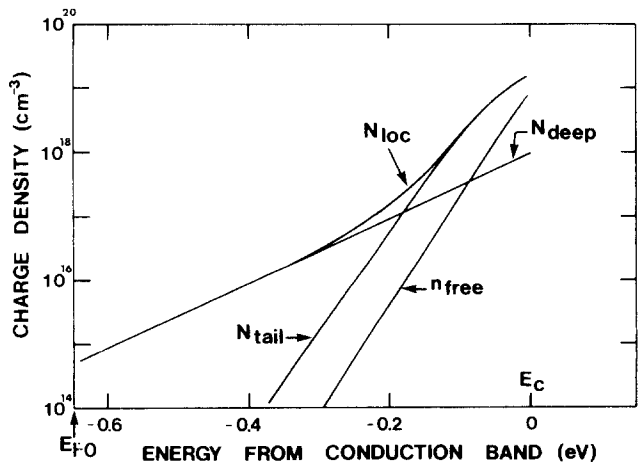


FIG. 4. Charge distribution in an a -Si FET as a function of Fermi level position. Q_{tail} —charge in localized tail states; Q_{deep} —charge in deep localized states; Q_{mobile} —charge induced into the band.

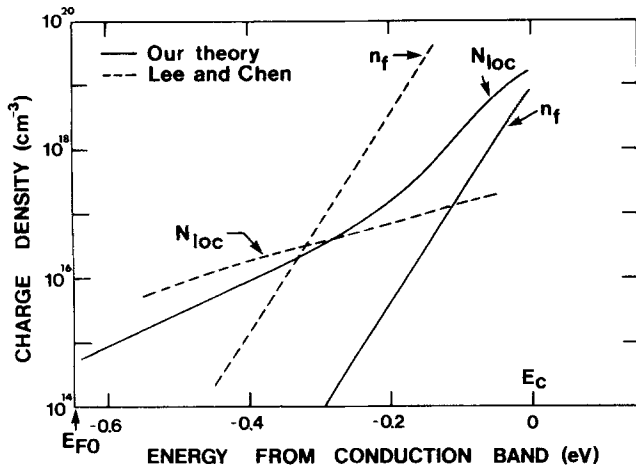


FIG. 5. Charge distribution in an *a*-Si FET as a function of the Fermi level position. Comparison between our theory (solid lines) and the Lee and Chen model (dashed lines).

the model developed in Ref. 15 is in the value of the free charge Q_F where

$$Q_F = qn_i \exp \Delta E_F / kT. \quad (17)$$

Here, ΔE_F is the Fermi level shift and n_i the intrinsic concentration as given by

$$n_i = \sqrt{N_c N_v} \exp\left(\frac{-E_g}{2kT}\right), \quad (18)$$

where N_c and N_v are effective densities of states in the conduction and valence band, respectively, and E_g is the energy gap. In Ref. 15 n_i was not calculated using Eq. (18) but rather was considered to be an adjustable parameter determined from the dark conductivity. However, the dark conductivity is determined by the position of the Fermi level which even in undoped amorphous silicon is above midgap. The approach used in Ref. 15 may lead to a large overestimate of the free carrier charge, as can be seen from Fig. 5. In order to explain the observed values of the device conductance they then had to assume an electron band mobility of $0.1 \text{ cm}^2/\text{Vs}$, which would imply a mean free path λ smaller than an interatomic distance. It may be noted that typical values of the drift mobility¹⁶ in *a*-Si are in the range 0.01 – $0.1 \text{ cm}^2/\text{Vs}$, much lower than the band mobility because of the large ratio of trapped to free carriers.¹²

IV. FIELD AND CHARGE DISTRIBUTION IN THE CHANNEL

We first find the electric field distribution in the channel by solving Poisson's equation

$$\frac{dF}{dx} = \frac{\rho}{\epsilon}, \quad (19)$$

where F is the electric field, x the space coordinate (perpendicular to the gate), ϵ the dielectric permittivity, and ρ the space-charge density:

$$\rho = -q(N_{\text{loc}} + n). \quad (20)$$

Here N_{loc} is the concentration of electrons in localized states,

$$n = N_c \exp\left(-\frac{(E_{F0} - qV)}{kT}\right), \quad (21)$$

is the concentration of mobile carriers in the conduction band. Here,

$$E_{F0} = E_c(\text{bulk}) - E_F, \quad (22)$$

and E_F is the equilibrium Fermi level position and V the potential (see Fig. 6).

Using the relationship between the electric field and potential V

$$F = -dV/dx, \quad (23)$$

we find

$$F^2 = \frac{2}{\epsilon} \int_0^V |\rho(V')| dV'. \quad (24)$$

We have chosen the reference potential ($V = 0$) to be the conduction band edge far from the channel.

The electric field at the interface is related to the total charge in the channel

$$F(0) = Q/\epsilon, \quad (25)$$

where

$$Q = Q_{\text{loc}} + q n_s$$

and

$$Q_{\text{loc}} = q \int_0^\infty N_{\text{loc}} dx, \quad (26)$$

where n_s is the sheet concentration of the free electrons in the channel. The value of n_s may be found as

$$n_s = n(0) d_{\text{eff}}, \quad (27)$$

where d_{eff} is the effective channel width given by

$$d_{\text{eff}} = kT/qF(0). \quad (28)$$

From Eqs. (21), (27), and (28) we find

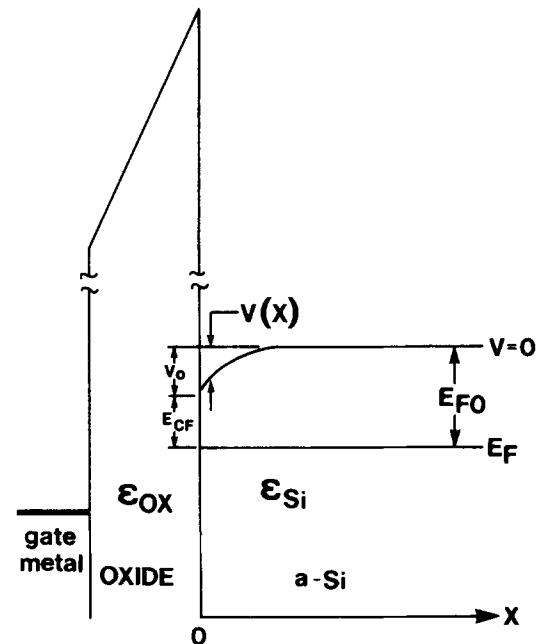


FIG. 6. Band diagram of an amorphous silicon TFT.

$$n_s = \frac{N_c kT}{qF(0)} \exp\left(\frac{-E_{cf}}{kT}\right). \quad (29)$$

From Eqs. (24) and (25) we find

$$Q = \left\{ 2\epsilon q \int_0^{V_0} [N_{\text{loc}}(V) + n(V)] dV \right\}^{1/2}, \quad (30)$$

where V_0 is the surface band bending (see Fig. 6).

The ratio of the mobile sheet charge to total sheet charge is given by

$$\frac{qn_s}{O} = \frac{N_c \exp\left(\frac{-E_{cf}}{kT}\right)}{\epsilon F^2(0)} kT, \quad (31)$$

and from Eq. (24)

$$\frac{qn_s}{Q} = \frac{N_c \exp\left(\frac{-E_{cF}}{kT}\right) kT}{2q \int_0^{V_0} [n(V) + N_{loc}(V)] dV}. \quad (32)$$

This ratio is equal to the ratio of the field-effect mobility μ_{fet} to the band mobility μ_0 .

Using the expressions for N_{loc} derived in Sec. III we may now find Q and n_s and use the results to determine the characteristics of α -Si TFT's.

V. GENERAL EXPRESSIONS FOR CURRENT-VOLTAGE AND CAPACITANCE-VOLTAGE CHARACTERISTICS OF *a*-Si TFT's

The current voltage characteristics may be found using the gradual channel approximation (see Fig. 7). The incremental voltage drop dV_{ch} in the channel is given by

$$dV_{\text{ch}} = IdR, \quad (33)$$

where I is the current and

$$dR = dy/q\mu_0 n_s W. \quad (34)$$

Here, y is the space coordinate in the direction parallel to the gate contact, μ_0 the band mobility, and W the gate width. Substituting Eq. (34) into Eq. (33) and integrating with respect to y from 0 to L (see Fig. 7) gate, where L is the gate length, we find

$$I_{ds} = \frac{q\mu_0 W}{L} \int_0^{V'_{ds}} n_s(V_{ch}) dV_{ch}, \quad (35)$$

where V'_{ds} is the total voltage drop across the channel. The drain-to-source voltage V_{ds} is given by

$$V_{ds} = V'_{ds} + I_{ds}(R_s + R_d), \quad (36)$$

where R_s and R_d are the series source and drain resistances.

The channel capacitance C_{ch} per unit gate area may be found as

$$C_{\text{ch}} \approx \frac{dQ}{dV_0}. \quad (37)$$

Using Eq. (30), Eq. (37) may be rewritten as

$$C_{\text{ch}} = \frac{\epsilon q [N_{\text{loc}}(V_0) + n(V_0)]}{Q}. \quad (38)$$

In Secs. VI and VII we will evaluate the integral in Eq. (35) and use Eqs. (35) and (38) in order to determine the I - V and C - V characteristics of a -Si TFT's both below and above threshold.

VI. CHARACTERISTICS OF *a*-Si TFT's BELOW THRESHOLD

As has been shown in Sec. III the density of the localized charge below threshold (when the electron Fermi level is in the deep localized states) is given by Eq. (8):

$$qN_{\text{deep}} = \frac{g_F k T \pi}{\sin(\pi T/T_c)} \quad (8)$$

(the contribution from the tail states in this region may be neglected). When the characteristic energy T_2 of the exponential variation of the density of the localized states is much larger than T Eq. (8) reduces to the zero temperature statistics expression:

$$qN_{\text{deep}} = g_F k T_2; \quad (39)$$

i.e., the charge is localized within the band of kT_2 below the Fermi level. Integration of Eq. (24) assuming $N_{\text{loc}} = N_{\text{deep}} \gg n$ for a practical case of a relatively large surface band bending $V_0 \gg kT_2/q$ yields

$$F(0) = \left(\frac{2\pi k T k T_2 g_{F0}}{\epsilon q \sin(\pi T/T_2)} \right)^{1/2} \exp\left(\frac{qV_0}{2kT_2} \right). \quad (40)$$

Here, g_{F0} is the density of the acceptor states at $E = E_{F0}$ so that

$$g_F = g_{F(0)} \exp\left(\frac{qV}{kT_s}\right). \quad (41)$$

Using Eqs. (30) and (40) we find

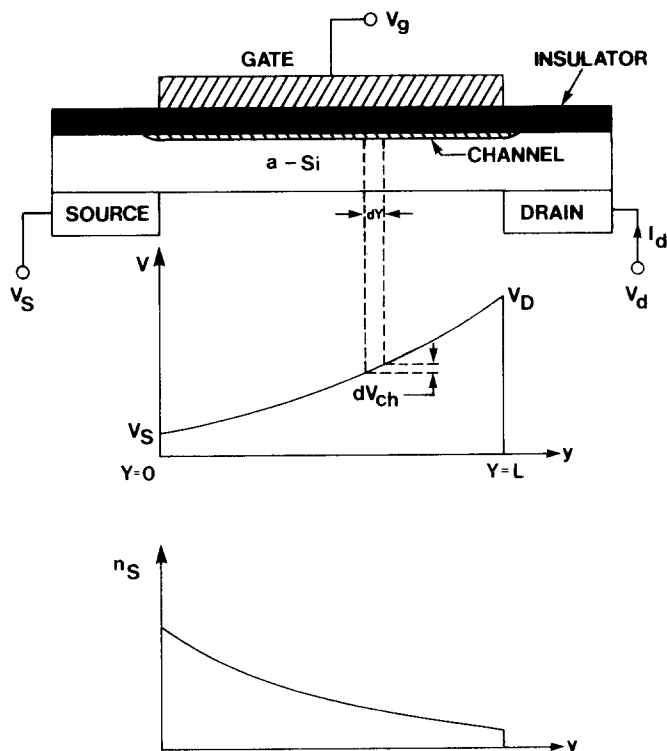


FIG. 7. Schematic diagram of an amorphous silicon TFT.

$$qn_s = qN_c \exp\left(\frac{-E_{F0}}{kT}\right) \left(\frac{\epsilon T \sin(\pi T/T_2)}{2\pi g_{F0} q T_2}\right)^{1/2} \exp(\beta V_0), \quad (42)$$

where

$$\beta = \left(\frac{1}{kT} - \frac{1}{2kT_2}\right)q, \quad (43)$$

and

$$E_{F0} = qV_0 + E_{cF} \quad (44)$$

(see Fig. 6). The ratio of qn_s/Q_{loc} is found from Eq. (31) to be

$$\frac{qn_s}{Q_{loc}} = \frac{qN_c \exp\left(\frac{-E_{F0}}{kT}\right)}{2\pi g_{F0} kT_2} \exp\left[qV_0\left(\frac{1}{kT} - \frac{1}{kT_2}\right)\right]. \quad (45)$$

Here, we have taken into account that in the below threshold regime $Q \simeq Q_{loc} \gg qn_s$.

The induced localized charge is given by

$$Q_{loc}(y) = C_{ox} V_{g1}(y), \quad (46)$$

where

$$V_{g1}(y) = V_g - V_{FB} - V_{ch}(y) \quad (47)$$

Here, V_g is the gate voltage with respect to the source side of the gate, V_{FB} the effective flat band voltage, and $V_{ch}(y)$ the channel potential.

The insulator capacitance per unit area C_{ox} is given by

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}, \quad (48)$$

and is much smaller than the channel capacitance per unit area C_{ch} [see Eq. (38)]. Here, ϵ_{ox} is the dielectric constant of the insulator layer and t_{ox} the thickness of the insulator layer. Hence, the voltage drop V_0 is much smaller than V_{g1} and V_{ch} is determined primarily by the drain-to-source potential. In this case, equating the localized charge given by Eqs. (29), (40), and (46) we find the band bending potential at the interface between the insulator and the channel:

$$V_0 = \frac{2kT_2}{q} \ln\left(\frac{C_{ox} V_{g1} [q \sin(\pi T/T_2)]^{1/2}}{\sqrt{2\pi\epsilon kT_2 kT g_{F0}}}\right). \quad (49)$$

The induced charge of the free electrons in the conduction band may be now found from Eqs. (42) and (49):

$$n_s = f(T, T_2) (C_{ox} V_{g1})^{\frac{2T_2}{T} - 1}, \quad (50)$$

where

$$f(T, T_2) = N_c \exp\left(\frac{-E_{F0}}{kT}\right) \frac{kT\epsilon}{q} \left(\frac{q \sin(\pi T/T_2)}{2\pi\epsilon kT_2 kT g_{F0}}\right)^{T_2/T}. \quad (51)$$

The I - V characteristics are found by substituting Eq. (50) for qn_s into Eq. (35) to yield

$$I_{ds} = \frac{q\mu_0 WT}{2T_2 L} f(T, T_2) C_{ox}^{\left(\frac{2T_2}{T} - 1\right)} \times \left[(V_g - V_{FB})^{\frac{2T_2}{T}} - (V_g - V_{FB} - V'_{ds})^{\frac{2T_2}{T}} \right]. \quad (52)$$

At small values of $V_{ds} \ll V_g - V_{FB}$ Eq. (52) may be simplified to

$$I_{ds} \approx \frac{q\mu_0 WT}{L} f(T, T_2) [C_{ox} (V_g - V_{FB})]^{\left(\frac{2T_2}{T} - 1\right)} V'_{ds}. \quad (53)$$

At

$$V'_{ds} \geq V_{ds}^{\text{sat}} = V_g - V_{FB}, \quad (54)$$

the current saturates at

$$I_{\text{sat}} = \frac{q\mu_0 WT}{2T_2 L} f(T, T_2) C_{ox}^{\left(\frac{2T_2}{T} - 1\right)} (V_g - V_{FB})^{\frac{2T_2}{T}}. \quad (55)$$

The voltage dependence of the current predicted by Eqs. (52) and (55) is in agreement with the analysis given in Ref. 17. However, our equations also reveal the temperature dependence of the subthreshold current and its dependence on the insulator capacitance.

Equations (52)–(55) are in good agreement with numerous experimental results.^{18–24} As an example we show in Fig. 8 $\log I_{ds}$ vs $\log V_g$ for data in the subthreshold regime measured by Thompson *et al.*¹⁸ As expected from Eq. (55) this plot is indeed linear and the slope of 6.9 corresponds to a T_2 of 1035 K in good agreement with the density of states spectrum for amorphous silicon.

We may note that in applying Eq. (55) to experimental data we have assumed that the flat band voltage is equal to zero. In fact, this flat band voltage has to be determined from the experimental data and the value for T_2 derived using Eq. (55) is dependent on this voltage. This will be further discussed in a later publication.²⁵ We have also checked that Eqs. (52)–(55) are in good agreement with the current voltage characteristics of a -Si TFT's in the subthreshold regime reported in Refs. 18–23. From the data given in Refs. 18–23 we find that kT_2/q varies from 80 to 100 mV.

In the model presented above we have ignored the off current. In an ideal case this off current I_{off} is given by

$$I_{\text{off}} \approx \frac{\sigma_D t W V'_{ds}}{L}, \quad (56)$$

where σ_D the dark conductivity of the a -Si and t the thickness of the a -Si film. This, however, assumes that there is no charge accumulation at the silicon-substrate interface. As

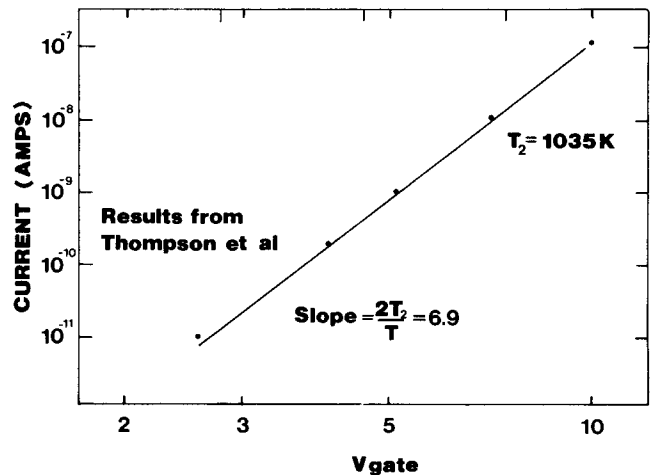


FIG. 8. I - V characteristic of an a -Si TFT in the below threshold regime from Thompson *et al.*

we find from our computer simulation and pointed out in Ref. 9, any band bending which increases the electron concentration in the a -Si at the interface away from the gate will drastically increase I_{off} . To some extent we could incorporate these effects into our model by adjusting V_{FB} and introducing a suitable shunt resistance. It may be noted that any band bending at the silicon-substrate interface will only be very significant in the below threshold regime.

The capacitance-voltage characteristic of the α -Si TFT at zero drain-to-source bias, may be found from

$$C = \frac{C_{\text{ox}} C_{\text{ch}}}{C_{\text{ox}} + C_{\text{ch}}}; \quad (57)$$

the capacitance C_{ch} may be found from Eqs. (8), (26), (38), (40), and (41), taking into account that $N_{\text{deep}} \gg n$

$$C_{\text{ch}} = \frac{C_{\text{ox}} q V_{g1}}{2kT_2} \gg C_{\text{ox}}. \quad (58)$$

This leads to

$$C \approx C_{\text{ox}} \left(1 - \frac{2kT_2}{qV_{g1}} \right). \quad (59)$$

Hence, we predict that the TFT capacitance in the subthreshold regime is a linear decreasing junction of $1/V_{g1}$ with a slope of $2kT_2/q$ and an intercept of C_{ox} .

When the band bending voltage V_0 becomes larger than $E_{F0} - E_{ct}$, where E_{ct} is the energy width of the tail acceptor states the device switches to what we call the "above threshold" regime. As can be found from Eq. (49) this occurs when

$$V_G = V_{TS} \equiv \frac{2\pi\epsilon k T_2 k T g_{F0}}{C_{\text{ox}} [q \sin(\pi T/T_2)]^{1/2}} \exp\left(\frac{E_{cF}}{2kT_2}\right) + V_{\text{FB}}. \quad (60)$$

For the values of the parameters given in Table I and for room temperature (300 K) we find

$$V_{TS} \approx 5V + V_{\text{FB}}.$$

This assumes, however, that the amorphous silicon layer is undoped. Doping shifts the bulk Fermi level position E_{F0} towards the conduction band edge such that

$$E_{F0}(N_D^+) \approx E_{F0}(0) + kT \ln\left(\frac{qN_D^+}{g_{F0} k T_2}\right). \quad (61)$$

This increases the off conductance as

$$\sigma_{\text{off}} = 1/q \mu_0 n_{\text{off}}, \quad (62)$$

where

$$n_{\text{off}} = N_c \exp\left(\frac{E_c - E_{F0}}{kT}\right), \quad (63)$$

and will also reduce the threshold voltage. The calculated dependencies of σ_{off} and V_{TS} vs $E_c - E_{F0}$ are shown in Fig. 9. As can be seen from the figure, doping can reduce the value of V_{TS} to a fraction of a volt for an insulator thickness of $0.25 \mu\text{m}$. Thus, the value of the threshold voltage can vary from a fraction of a volt to several volts depending on doping concentration and insulator thickness.

However, doping also leads to a decrease in the on to off ratio of the device. This is given by

$$\frac{I_{\text{on}}}{I_{\text{off}}} = \frac{j_{\text{on}} L}{V_{\text{on}} \sigma_{\text{off}} t}, \quad (64)$$

where L is the gate length, t the thickness of the a -Si layer, j_{on} the on current per unit gate width, and V_{on} the on voltage.

Assuming j_{on} is approximately $2 \times 10^{-7} \text{ A}/\mu\text{m}$ and V_{on} about 20 V (which is what may be expected according to the calculation given in Sec. VII), $L = 10 \mu\text{m}$ and $t = 0.4 \mu\text{m}$, we find from Eq. (64) that

$$\frac{I_{\text{on}}}{I_{\text{off}}} \sim \frac{0.025}{\sigma_{\text{off}} (\Omega \text{ cm})^{-1}}. \quad (65)$$

Thus, Fig. 9 may be used to establish the trade off between the reduction of the threshold voltage and the lower on to off current ratio that can be achieved as result of doping the amorphous silicon.

VII. TRANSITION TO THE ABOVE THRESHOLD REGIME AND a -Si TFT's CHARACTERISTICS ABOVE THRESHOLD

In the above threshold regime most of the localized charge goes into the tail states. However, in our analysis of the above threshold regime we also take into account the deep localized states in order to follow the transition from the below threshold to the above threshold regime. Hence, the results obtained in this section describe both regimes. Below threshold the analytical equations derived in Sec. VI

TABLE I. List of parameters used in all calculations in this paper unless otherwise specified.

$g_{F0} = 5 \times 10^{15} \text{ cm}^{-3}/\text{eV}$
$g_{cd} = 9.6 \times 10^{18} \text{ cm}^{-3}/\text{eV}$
$T_1 = 260 \text{ K } (kT_1/q = 22.4 \text{ mV})$
$T_2 = 1000 \text{ K } (kT_2/q = 86 \text{ mV})$
$g_{ct} = 10^{21} \text{ cm}^{-3} \text{ eV}^{-1}$
$E_{ct} = 0.142 \text{ mV}$
$t_{\text{ox}} = 0.25 \mu\text{m}$
$\epsilon_{\text{ox}} = 3.45 \times 10^{-11} \text{ F/m}$
$N_c = 10^{25} \text{ m}^{-3}$
$\epsilon_{\text{si}} = 10^{-10} \text{ F/m}$

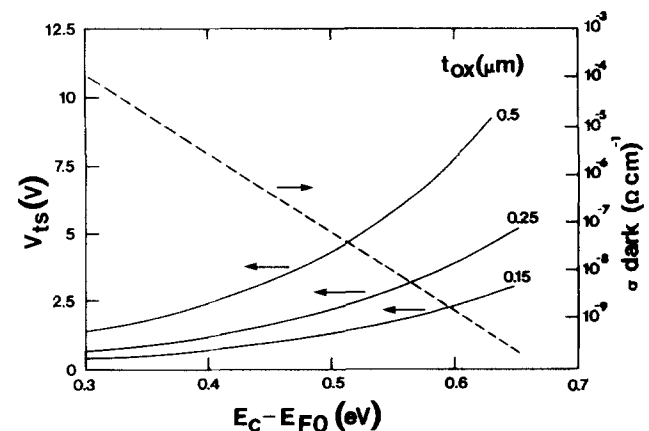


FIG. 9. Critical voltage required to fill the tail states vs the equilibrium position of the Fermi level. Also shown is the dark conductivity.

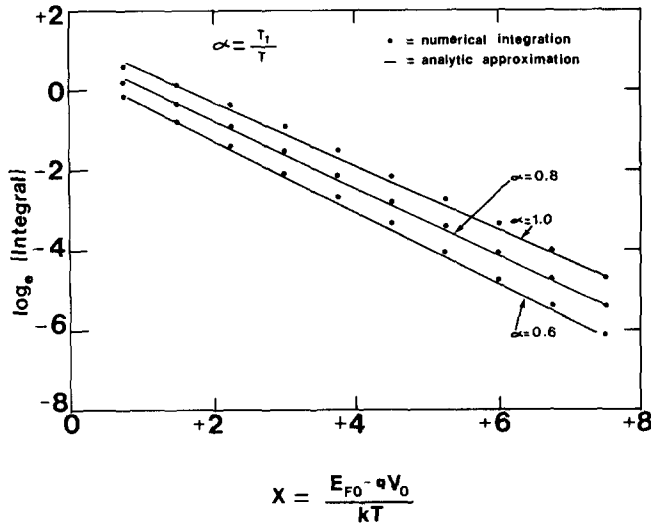


FIG. 10. Comparison between the numerical calculation and the interpolation formula for the integral determining the contribution of the tail states to the total induced charge in the channel.

are quite adequate, although for the above threshold regime a new approach has to be used.

The induced charge in the channel may be presented as

$$qN_{\text{ind}} \equiv CV_{g1} = qn_s + Q_{\text{deep}} + Q_{\text{tail}}. \quad (66)$$

Equation (66) may be rewritten as

$$qN_{\text{ind}} = \left\{ 2\epsilon q \int_0^{V_0} [n(V) + N_{\text{tail}}(V) + N_{\text{deep}}(V)] dV \right\}^{1/2} \quad (67)$$

[see Eq. (30)]. Here,

$$n(V) = N_c \left[\exp(qV - E_{F0})/kT \right], \quad (68)$$

$$N_{\text{tail}} = \frac{1}{2} g_{ct} kT_1 \left\{ 2 \ln \left[\exp \left(\frac{E_{F0} - qV}{kT} \right) + 1 \right] \right\}^{T_1/T} \times \exp \left(\frac{qV - E_{F0}}{kT} \right), \quad (69)$$

and

$$N_{\text{deep}} = g_{F0} kT \frac{\pi}{\sin(\pi T/T_2)} \exp \left(\frac{qV}{kT_2} \right) \quad (70)$$

[see Eqs. (4), (8), and (12)]. The integral in the right hand side of Eq. (67) may be presented as the sum of three integrals. Two of them

$$\int_0^{V_0} n(V) dV \quad \text{and} \quad \int_0^{V_0} N_{\text{deep}}(V) dV,$$

are easily evaluated analytically. The third integral

$$\int_0^{V_0} N_{\text{tail}}(V) dV = \frac{1}{2} g_{ct} kT \frac{kT_1}{q} I(\alpha, x_0), \quad (71)$$

where

$$I(\alpha, x_0) \approx \int_0^{x_0} [2 \ln(e^{\alpha x} + 1)]^\alpha e^{-x} dx, \quad (72)$$

has to be evaluated numerically. Here, $x_0 = (E_{F0} - qV_0)/kT$ and $\alpha = T_1/T$.

In Fig. 10 the results of the numerical integration are compared with a simple interpolation formula given by

$$I(\alpha, x_0) = \exp[A(\alpha) + B(\alpha)x_0], \quad (73)$$

where

$$A(\alpha) = 1.92\alpha - 0.595, \quad (74)$$

$$B(\alpha) = 0.225\alpha - 1.03. \quad (75)$$

As can be seen from the figure, this interpolation formula is in very good agreement with the numerical integration and, hence, may be used in our analytical theory.

From Eqs. (67)–(71) we obtain

$$qN_{\text{ind}} = \left\{ 2\epsilon q \left[\frac{PkT_2}{q} \exp \left(\frac{qV_0}{kT_2} \right) + G(V_0) + \frac{kT}{q} N_c \exp(-x_0) \right] \right\}^{1/2}, \quad (76)$$

where

$$P = \frac{\pi g_{F0} kT}{q \sin(\pi T/T_2)}, \quad (77)$$

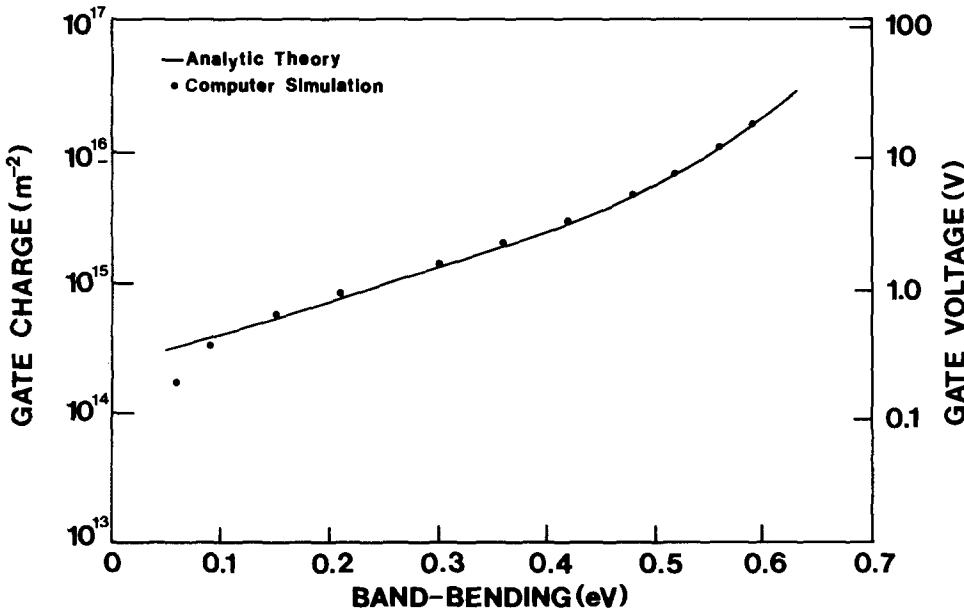


FIG. 11. Gate charge/voltage as a function of band bending for both our analytical theory and the complete computer simulation.

$$G = \frac{1}{2} g_{cr} \frac{kT_1}{q} \frac{kT}{q} I(\alpha, x_0). \quad (78)$$

Once the dependence of qN_{ind} vs V_0 is calculated where

$$qN_{ind} \approx C_{ox} V_{g1}. \quad (79)$$

n_s , μ_{fet}/μ_0 , C_{ch} and other characteristics of the TFT structure may be found using the equations given in Sec. V. The values of parameters given in Table I are used in this calculation.

The calculated dependence of V_0 vs N_{ind} is shown in Fig. 11. The dots shown in the figure indicate the results of the computer simulation based on the numerical integration of the complete system of transport equations. The numerical model is based on our amorphous silicon solar cell model.¹¹⁻¹⁴ Except for the region where the band bending V_0 is smaller than several kT_2/q (in this region the analytical model does not apply) the agreement between the analytical and computer simulation is excellent. Notice that in the below threshold regime V_0 and n_s increase rapidly with V_{g1} and that the increase is much more gradual at large gate voltages.

The effective field-effect mobility

$$\mu_{fet} = \mu_0 \frac{n_s}{N_{ind}}, \quad (80)$$

is one of the most important device parameters. The calculated curves of $\log \mu_{fet}/\mu_0$ are shown in Fig. 12 as functions of the gate voltage V_{g1} for different sample temperatures. As seen from the figure at small gate voltages (in the below threshold regime) $\mu_{fet}/\mu_0 \ll 1$ and this ratio is a strong function of both gate voltage and temperature.

The dependences of μ_{fet}/μ_0 vs V_{g1} for different values of T_1 are shown in Fig. 13. A change in T_1 from 220 to 300 K leads to a considerable change in μ_{fet} at high gate voltages. Of course the values of μ_{fet} at low gate voltages (in the below threshold regime) are independent of T_1 . Hence, a decrease in the density of the localized tail states increases the field-effect mobility above threshold whereas a decrease in the

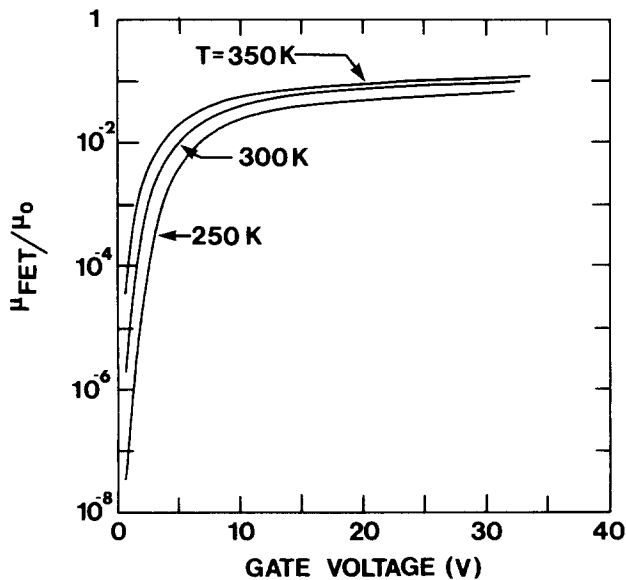


FIG. 12. Ratio of field-effect to band mobility as a function of gate voltage for varying temperatures (log scale).

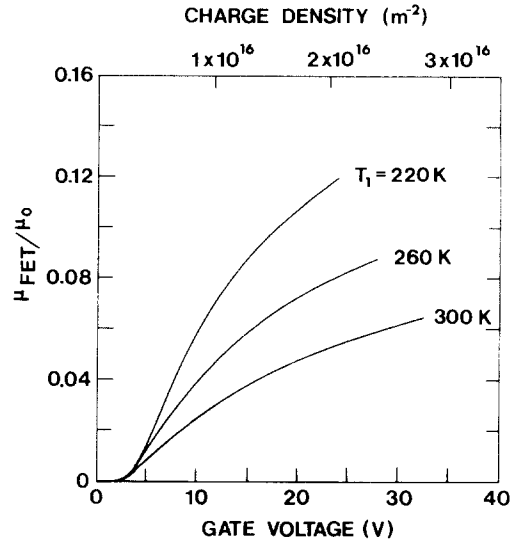


FIG. 13. Ratio of field-effect to band mobility as a function of the gate voltage for varying characteristic energy slopes kT_1 of the exponentially varying tail states.

density of the deep localized states primarily decreases the threshold voltage.

In Fig. 14 the dependences of μ_{fet}/μ_0 vs V_{g1} are plotted in a linear scale. As can be seen from this figure this dependence may be quite accurately approximated as

$$\frac{\mu_{fet}}{\mu_0} = K'(V_{g1} - V_T)^{1/2}, \quad (81)$$

where V_T is the effective threshold voltage. For comparison, the field-effect mobility in the crystalline MOSFET's for the above threshold regime is independent of the gate voltage. At 300 K we find $K' \approx 0.018$ and $V_T \approx 4$ V (see Fig. 14).

The channel capacitance C_{ch} vs gate voltage characteristics are calculated using Eq. (38) (see Fig. 15). The slope of Fig. 15 at low gate voltages is in good agreement with Eq. (58). As can be seen from the figure the assumption $C_{ox} \ll C_{ch}$ used in this paper is valid for all relevant gate voltages.

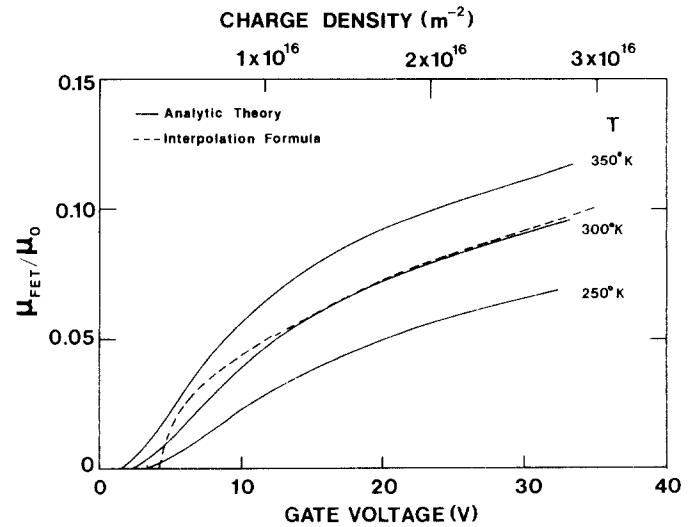


FIG. 14. Ratio of field-effect to band mobility as a function of gate voltage for varying temperatures (linear scale). Dotted line shows an interpolation for the above threshold regime at 300 K.

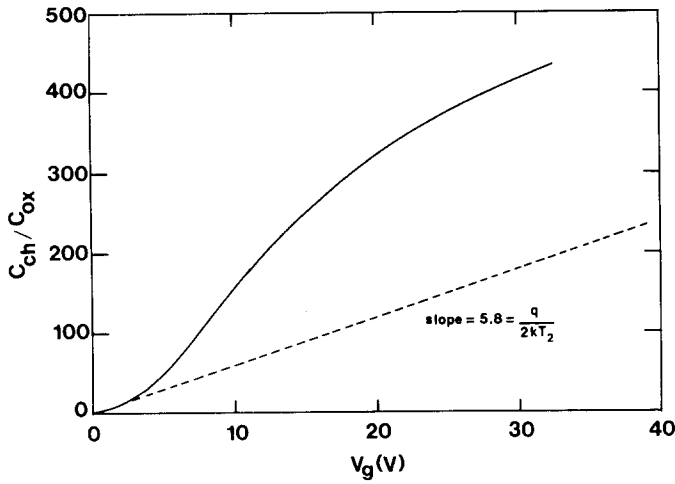


FIG. 15. Ratio of channel to insulator capacitance as a function of gate voltage.

We can now calculate the current-voltage characteristics of an α -Si TFT using Eq. (35). When the drain-to-source voltage is applied V_{g1} in Eq. (81) should be substituted by

$$V_1 = V_g - V(x), \quad (82)$$

where $V(x)$ is the channel potential and

$$\mu_{\text{eff}} = \mu_0 K' [V_g - V(x) - V_T]^{1/2}. \quad (83)$$

Substitution of Eq. (83) using Eq. (35) yields

$$I_{ds} = \frac{\mu_0 W C_{ox} K'}{L} \int_0^{V'_{ds}} (V_g' - V)^{3/2} dV, \quad (84)$$

where

$$V_g' = V_g - V_T - I_{ds} R_s, \quad (85)$$

$$V'_{ds} = V_{ds} - I_{ds} (R_s + R_d), \quad (86)$$

and R_s and R_d the source and drain series resistances. Equation (84) is only valid for drain-to-source voltages smaller than the saturation voltage. From Eq. (84) we find

$$I_{ds} = \frac{\mu_0 W K' C_{ox}}{2.5L} [(V_g')^{2.5} - (V_g' - V'_{ds})^{2.5}]. \quad (87)$$

At small drain-to-source voltages $V'_{ds} \ll V_g^1$ and Eq. (87) may be simplified to

$$I_{ds} \sim \frac{\mu_0 W K' C_{ox} (V_g')^{1.5}}{L} V'_{ds}. \quad (88)$$

At drain-to-source voltages larger than the saturation drain-to-source voltage, i.e., for

$$V'_{ds} > V_g^1, \quad (89)$$

the drain-to-source current is equal to

$$I_{ds}^{\text{sat}} = \frac{\mu_0 W K' C_{ox}}{2.5L} (V_g')^{2.5}. \quad (90)$$

This may be compared to the standard "square law" dependence

$$I_{ds}^{\text{sat}} \sim (V_g')^2, \quad (91)$$

for crystalline MOSFET's.

The current-voltage characteristics calculated for the values of the parameters given in Table I and for

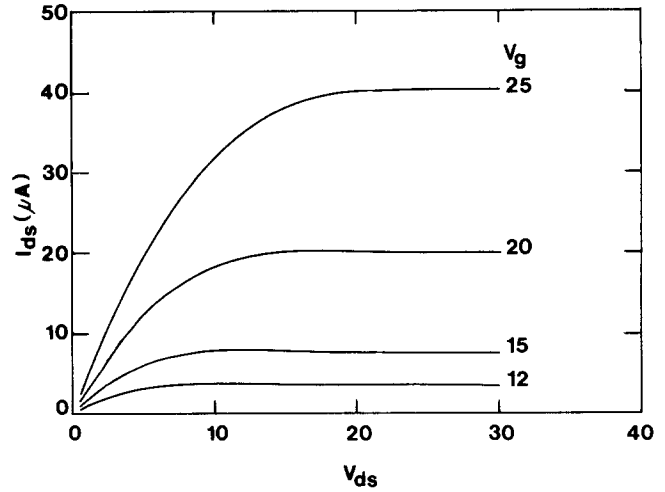


FIG. 16. Calculated current-voltage characteristics for varying gate voltages, and zero source and drains series resistances.

$R_s = R_d = 0$ are shown in Fig. 16. As can be seen from the figure the values of the on current are of the order of 2×10^{-7} A/ μm gate width for a 10- μm gate device at $V_g \approx 25$ V.

In Fig. 17 the predicted dependence of the transconductance g_m^s in the saturation regime as given by

$$g_m^s = \frac{C_1 (V_g - V_T)^{1.5}}{1 + C_1 (V_g - V_T)^{1.5} R_s}, \quad (92)$$

is compared with experimental results reported by Ast.²⁶ The deduced values of C_1 , V_T , and R_s are 0.8×10^{-7} mho/ $V^{1.5}$, 7 V, and 1.68 M Ω , respectively. This corresponds to the experimental field-effect mobility of 0.2 cm^2/Vs at $V_g = 20$ V. Also shown in Fig. 17 for comparison are g_m vs V_g curves for $R_s = 0.84$ M Ω and $R_s = 0.42$ M Ω . These results clearly illustrate the crucial role played by the source series resistance in limiting the transconductance and the on current of an amorphous silicon TFT.

In all our calculations in this paper we have assumed no interface states at either the insulator-semiconductor interface or at the back surface of the active layer. As noted by Powell and Pritchard,⁸ surface states may considerably re-

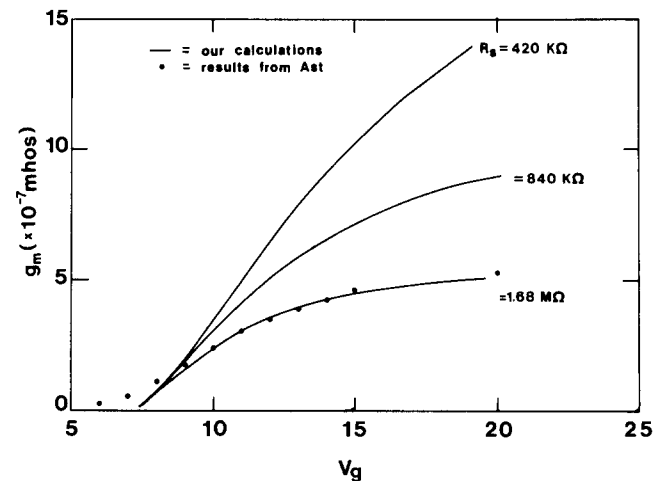


FIG. 17. Transconductance as a function of gate voltage for varying values of the source series resistance.

duce the field-effect conductance of amorphous silicon TFT's, especially in the subthreshold regime. In a later publication²⁵ we discuss in detail the effects of interface states on device performance.

VIII. CONCLUSION

Our results show that the characteristics of *a*-Si TFT's are primarily determined by the localized states in the energy gap of amorphous silicon. The distribution of the deep localized states determines the threshold voltage and the device behavior below threshold, whereas the distribution of the localized tail states controls the above threshold characteristics. The field-effect mobility is gate voltage dependent and may reach a significant (10%–15%) of the band mobility at high gate voltages. The source series resistance plays a very important role in limiting the performance of practical devices.

The validity of our analytical approach is confirmed by an accurate computer simulation based on the numerical integration of the complete set of the transport equations. Our results are also in good agreement with available experimental data. Finally, our theory of *a*-Si TFT's may be applied to the design of *a*-Si field-effect transistors and integrated circuits.

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